MODULE 07 SUMMARY

Trigonometric Solutions and Identities

Big Picture

Trigonometry is built on structure and repeated reasoning. By understanding unit circle patterns and quadrant behavior, we can determine exact values without a calculator. These exact values become the foundation for identities, equation solving, and modeling.

Essential Standard

Students will understand and apply trigonometric identities and advanced solution strategies, including unit circle reasoning, identity manipulation, equation solving in radians, and mathematical modeling through approximation methods.

Objective A — Exact Trigonometric Values

I can determine exact values of sine, cosine, and tangent for special angles on the unit circle and determine signs using quadrant structure.

Key Vocabulary

- Unit Circle
- Reference Angle
- Coterminal Angle
- Quadrant
- Radians

Core Structure

Cosine = x-coordinate

Sine = y-coordinate

Tangent = $\frac{\sin\theta}{\cos\theta}$

Cosecant = $\frac{1}{\sin\theta}$

Secant = $\frac{1}{\cos\theta}$

Cotangent = $\frac{1}{\tan\theta}$

Quadrant I Patterns (Memorize These Only)

$$\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$$

$$\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$$

$$\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$$

Values repeat across quadrants. Only the signs change.

Sign Reminder (ASTC Pattern)

- Quadrant I: All positive
- Quadrant II: Sine positive
- Quadrant III: Tangent positive
- Quadrant IV: Cosine positive

Key Reminders

- Reduce angles greater than 2π using coterminal angles.
- Negative angles rotate clockwise.
- Always determine quadrant before assigning sign.

Sample Questions

1. $\sin\left(\frac{3\pi}{4}\right)$

2. $\cos\left(-\frac{\pi}{6}\right)$

3. $\tan\left(\frac{5\pi}{3}\right)$

4. $\sin\left(\frac{19\pi}{6}\right)$

Workspace

Objective B — Double-Angle & Pythagorean Identities

I can apply double-angle identities and Pythagorean identities to simplify and rewrite trigonometric expressions while justifying each algebraic step.

Key Vocabulary

- Identity
- Double-Angle Identity
- Pythagorean Identity
- Equivalent Forms
- Algebraic Justification

Core Identities

$$\sin(2\theta) = 2\sin\theta\cos\theta$$

$$\cos(2\theta) = \cos^2\theta - \sin^2\theta$$

$$\cos(2\theta) = 2\cos^2\theta - 1$$

$$\cos(2\theta) = 1 - 2\sin^2\theta$$

$$\sin^2\theta + \cos^2\theta = 1$$

$$1 + \tan^2\theta = \sec^2\theta$$

$$1 + \cot^2\theta = \csc^2\theta$$

Structure Insights

- Cosine has three equivalent double-angle forms.
- Choose the form strategically based on what the expression contains.
- Pythagorean identities allow substitution between sine and cosine.
- Every step must be justified — identities rewrite, they do not solve.

Common Rewrites

Rewrite $\cos(2\theta)$ in terms of sine only:

$$\cos(2\theta) = 1 - 2\sin^2\theta$$

Rewrite $\cos(2\theta)$ in terms of cosine only:

$$\cos(2\theta) = 2\cos^2\theta - 1$$

Sample Questions

1. Evaluate $\sin(2\theta)$ if $\sin\theta = \frac{3}{5}$ and θ is in Quadrant I.

2. Rewrite $\cos(2\theta)$ in terms of sine only.

3. Simplify $\frac{1-\cos(2\theta)}{2}$.

4. Simplify $\frac{\sin^2\theta}{1-\cos^2\theta}$.

Workspace

Objective C — Half-Angle, Sum, and Difference Identities

I can apply half-angle formulas and sum/difference identities to rewrite and evaluate trigonometric expressions with precision, including correct sign determination.

Key Vocabulary

- Half-Angle Formula
- Sum Identity
- Difference Identity
- Reference Angle
- Sign Determination

Half-Angle Formulas

$$\sin\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1-\cos\theta}{2}}$$

$$\cos\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1+\cos\theta}{2}}$$

$$\tan\left(\frac{\theta}{2}\right) = \frac{\sin\theta}{1+\cos\theta}$$

Important: The $\pm$ depends on the quadrant of $\frac{\theta}{2}$.

Sum and Difference Identities

$$\sin(A+B) = \sin A\cos B + \cos A\sin B$$

$$\sin(A-B) = \sin A\cos B - \cos A\sin B$$

$$\cos(A+B) = \cos A\cos B - \sin A\sin B$$

$$\cos(A-B) = \cos A\cos B + \sin A\sin B$$

Structure Insights

- Half-angle formulas introduce square roots and ± signs.
- Quadrant determines the correct sign.
- Cosine switches signs between sum and difference forms.
- Careful algebra prevents sign errors.
- Fully simplify radicals when possible.

Sample Questions

1. Evaluate $\sin\left(\frac{\pi}{8}\right)$.

2. Evaluate $\cos(75^\circ)$.

3. Rewrite $\sin(45^\circ + 30^\circ)$.

4. Simplify $\cos(A-B) + \cos(A+B)$.

Workspace

Objective D — Solving Advanced Trigonometric Equations

I can solve advanced trigonometric equations in radians by isolating the trig function, identifying reference angles, determining all solutions in an interval, and writing general solutions when required.

Key Vocabulary

- Principal Solution
- General Solution
- Period
- Reference Angle
- Verification

Core Strategy

1. Isolate the trigonometric function.
2. Identify the reference angle using unit circle values.
3. Determine all solutions in the given interval.
4. Write the general solution using correct period notation (if required).
5. Verify solutions when necessary.

Important Reminders

- Include **all** quadrant solutions.
- Work in radians unless stated otherwise.
- Use correct period:
  - 2π for sine and cosine
  - π for tangent
- Do not stop after one solution.
- Check for extraneous solutions when squaring.

Sample Questions

1. Solve $\sin\theta = \frac{1}{2}$ for $0 \leq \theta < 2\pi$.

2. Solve $2\cos\theta - 1 = 0$.

3. Solve $\sec\theta = 2$.

4. Solve $2\sin^2\theta - 1 = 0$.

Workspace

Objective E — Riemann and Trapezoidal Approximations

I can approximate the area under a curve using left and right Riemann sums and trapezoidal sums from functions and intervals, and I can justify which method provides a better estimate.

Key Vocabulary

- Riemann Sum
- Left Endpoint Approximation
- Right Endpoint Approximation
- Trapezoidal Sum
- Subinterval Width (Δx)
- Approximation Error

Core Formulas

$$\text{Left Sum: } \sum f(x_{left})\Delta x$$

$$\text{Right Sum: } \sum f(x_{right})\Delta x$$

$$\text{Trapezoidal Sum: } \frac{\Delta x}{2}\left[f(x_0) + 2f(x_1) + \cdots + 2f(x_{n-1}) + f(x_n)\right]$$

Structure Insights

- If a function is increasing, left underestimates and right overestimates.
- If decreasing, the opposite occurs.
- Trapezoidal sums generally improve accuracy by averaging.
- More subintervals → better approximation.
- Always interpret area in context.

Applied Practice

1. Approximate the area under $f(x) = x^2 + 1$ on $[1, 5]$ using a left Riemann sum with 4 subintervals.

2. Approximate the area under $f(x) = \sqrt{x}$ on $[2, 6]$ using a right Riemann sum with 4 subintervals.

3. Approximate the area under $f(x) = \sin x$ on $\left[\frac{\pi}{4}, \frac{5\pi}{4}\right]$ using a trapezoidal sum with 4 subintervals.

4. For $f(x) = x^3 - 2x$ on $[-1, 3]$, compute both a left Riemann sum and a trapezoidal approximation using 4 subintervals. Which method appears more accurate? Justify mathematically.